Experiment 3: Antenna Characterization

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1 Aim of the Experiment

To experimentally characterize a monopole/dipole antenna by evaluating its resonant frequency, input impedance match, distance-dependent power fall-off, and far-field radiation patterns in both co-polarization and cross-polarization configurations:

1. Measuring Voltage Standing Wave Ratio (VSWR), reflection coefficient ($|\Gamma|$), and Return Loss (RL) to determine antenna resonance and impedance matching.
2. Investigating the distance dependence of received power in the far-field regime and evaluating compliance with the inverse-square propagation law.
3. Recording 360° angular beam patterns in E-plane and H-plane for both co-polarized and cross-polarized orientations.
4. Determining the Half-Power Beamwidth (HPBW) in principal planes and estimating antenna directivity (D).

2 Theory and Principles

2.1 Monopole Resonance and Input Impedance Matching

A monopole antenna of physical length L mounted above a conducting ground plane behaves as a half-wave dipole via image theory. Resonance occurs when the physical length equals a quarter-wavelength ($\lambda/4$):

$$L = \frac{\lambda}{4} = \frac{c}{4f_r} \implies f_r = \frac{c}{4L} \tag{1}$$

where $c = 3 \times 10^8\,\text{m/s}$ is the speed of light in vacuum and f_r is the resonant operating frequency.

When connected to a transmission line of characteristic impedance $Z_0 = 50\,\Omega$, an impedance mismatch between antenna input impedance Z_L and Z_0 induces a reflected wave. The voltage reflection coefficient Γ and VSWR are defined as:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{2}$$

$$\text{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \tag{3}$$

The corresponding Return Loss (RL) in decibels is:

$$\text{RL (dB)} = -20 \log_{10} |\Gamma| \tag{4}$$

At resonance, input reactance vanishes, minimizing VSWR ($\to 1$) and maximizing return loss.

2.2 Friis Transmission Formula and Distance Dependence

In free space, the power received (P_r) by an antenna at a separation distance R from a transmitting antenna of gain G_t radiating power P_t is governed by the Friis Transmission Formula:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 \tag{5}$$

Expressing power in logarithmic scale (dBm):

$$P_r\ (\text{dBm}) = P_t\ (\text{dBm}) + G_t\ (\text{dBi}) + G_r\ (\text{dBi}) + 20\log_{10}\left(\frac{\lambda}{4\pi}\right) - 20\log_{10} R \tag{6}$$

This establishes a theoretical $-20\,\text{dB}$/decade linear fall-off with respect to $20\log_{10} R$.

2.3 Radiation Pattern, Half-Power Beamwidth, and Directivity

The radiation pattern $P(\theta, \phi)$ represents the spatial distribution of radiated power.

- **E-Plane:** The plane containing the electric field vector $\vec{E}$ and the direction of maximum radiation.
- **H-Plane:** The plane containing the magnetic field vector $\vec{H}$ and the direction of maximum radiation (perpendicular to the E-plane).

The Half-Power Beamwidth (HPBW) is the angular width between the two points where received power drops by 3 dB relative to the peak:

$$P(\theta_{-3\text{dB}}) = P_{\max} - 3.0\,\text{dB} \tag{7}$$

Assuming a smooth main lobe, antenna directivity D can be approximated from principal plane beamwidths (θ_E and θ_H in radians):

$$D \approx \frac{4\pi}{\theta_E \theta_H} \implies D\ (\text{dBi}) = 10\log_{10} D \tag{8}$$

3 Observation Tables and Quantitative Analysis

3.1 Part A: Monopole Antenna Resonance and Impedance Match

Monopole Physical Length (L): 37.5 cm = 0.375 m

Table 1: VSWR, Reflection Coefficient, Return Loss, and Wavelength Measurements

Frequency f	$V_{\max}$ (mV)	$V_{\min}$ (mV)	**VSWR**	—Γ|	**Return Loss RL (dB)**	**Wavelength λ (m)**
200.0 MHz	412.0	384.0	**1.073**	**0.0352**	**29.12**	1.500
30.0 MHz	430.0	240.0	**1.792**	**0.2836**	**20.55**	10.000

Resonance Verification Calculations: The theoretical quarter-wave resonant frequency for $L = 0.375$ m is:

$$f_r = \frac{c}{4L} = \frac{3 \times 10^8}{4 \times 0.375} = \mathbf{200.0}\ \text{MHz}$$

At $f = 200$ MHz:

$$\text{VSWR} = \frac{412}{384} = \mathbf{1.073} \approx 1.07 \tag{9}$$

$$|\Gamma| = \frac{1.073 - 1}{1.073 + 1} = \frac{0.073}{2.073} = \mathbf{0.0352} \tag{10}$$

$$\text{RL} = -20 \log_{10}(0.0352) = \mathbf{29.12}\ \text{dB} \tag{11}$$

$$\lambda = \frac{3 \times 10^8}{200 \times 10^6} = 1.500\ \text{m} \implies \frac{\lambda}{4} = \mathbf{0.375}\ \text{m} = L \tag{12}$$

At $f = 30$ MHz, VSWR = 1.792 ($|\Gamma| = 0.2836$, RL = 20.55 dB), indicating significant off-resonance reactive mismatch.

3.2 Part B: Effect of Distance on Received Power

Configuration: Pre-amplifier ON, Attenuation set to 20 dB on VNA.

Table 2: Received power vs. antenna separation distance R

Distance R (cm)	**Distance R (m)**	$20 \log_{10} R$	**Received Power P_r (dBm)**
904.0	9.04	19.12	-46.80
744.0	7.44	17.43	-38.77
584.0	5.84	15.33	-29.77
424.0	4.24	12.55	-30.90
264.0	2.64	8.43	-26.96

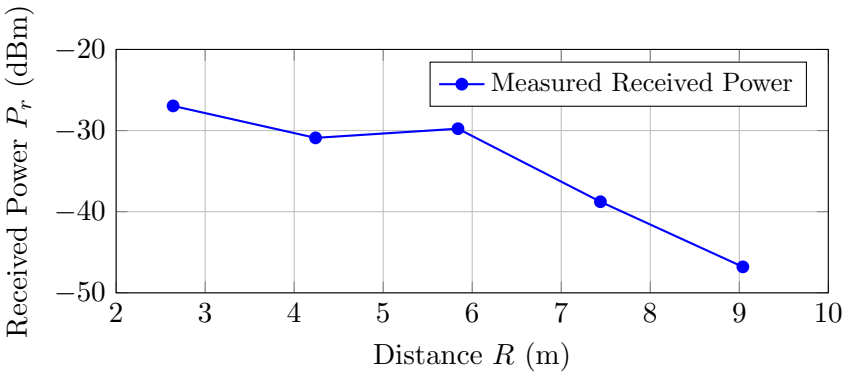


Figure 1: Received power P_r vs. physical separation distance R.

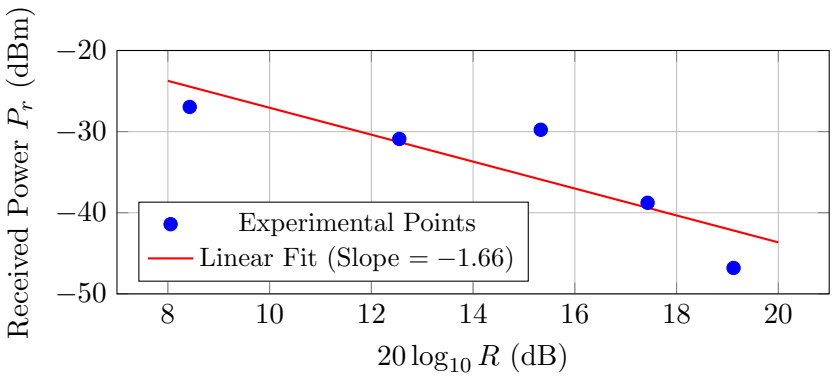


Figure 2: Linear regression of P_r vs. $20\log_{10} R$ validating the distance fall-off trend.

3.3 Part C: Co-Polarization Radiation Pattern Data

Table 3: Co-Polarization E-plane and H-plane Received Power vs. Angle θ

Angle θ (°)	**E-Plane (dBm)**	**H-Plane (dBm)**	**Angle θ (°)**	**E-Plane (dBm)**	**H-Plane (dBm)**
0	-27.40	-44.70	190	-52.76	-58.93
10	**-27.34 (Max)**	-44.66	200	-42.60	-51.51
20	-27.60	-43.91	210	-41.70	-50.41
30	-28.51	-42.93	220	-40.30	-50.90
40	-29.30	-42.75	230	-40.00	-52.88
50	-30.91	-42.05	240	-40.93	-53.50
60	-32.73	-41.69	250	-42.20	-53.54
70	-34.20	-40.93	260	-39.36	-54.15
80	-36.91	-40.88	270	-38.10	-50.95
90	-41.20	-41.03	280	-36.66	-44.40
100	-41.90	-42.90	290	-34.23	-41.64
110	-40.50	-51.52	300	-32.50	-40.30
120	-41.70	-52.30	310	-31.43	-38.88
130	-39.90	-51.20	320	-30.40	**-38.49 (Max)**
140	-39.12	-46.91	330	-30.18	-39.70
150	-41.80	-47.70	340	-28.77	-41.20
160	-46.50	-47.57	350	-28.17	-43.17
170	-46.60	-49.41	360	-27.62	-44.74
180	-46.10	-53.76			

3.4 Part D: Cross-Polarization Radiation Pattern Data

Table 4: Cross-Polarization E-plane and H-plane Received Power vs. Angle θ

Angle θ (°)	**E-Plane (dBm)**	**H-Plane (dBm)**	**Angle θ (°)**	**E-Plane (dBm)**	**H-Plane (dBm)**
0	-62.68	-27.76	200	-56.67	-41.84
20	-50.50	-28.32	220	-56.40	-44.70
40	-48.06	-30.50	240	-52.70	-45.67
60	-48.62	-32.12	260	-43.11	-57.25
80	-48.80	-32.92	280	-43.00	-40.80
100	-52.88	-33.53	300	-44.15	-32.50
120	-49.37	-39.61	320	-47.81	-29.51
140	-43.10	-41.41	340	-67.13	-27.66
160	-54.50	-47.67	360	-62.68	-27.76
180	-58.50	-40.89			

4 Co-Polarization Radiation Pattern

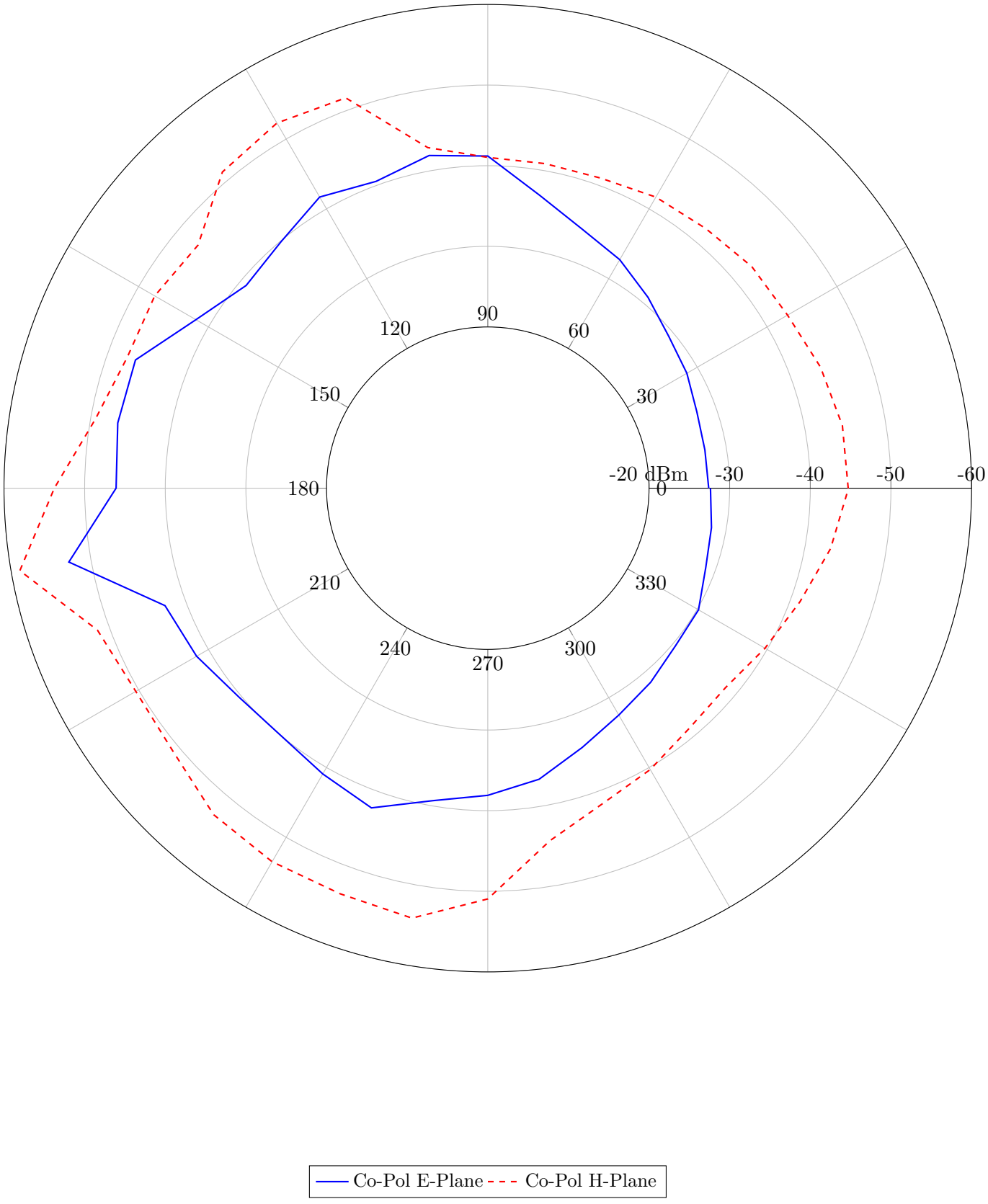


Figure 3: 2D Polar radiation pattern plot for Co-Polarized E-plane and H-plane configurations.

5 Cross-Polarization Radiation Pattern

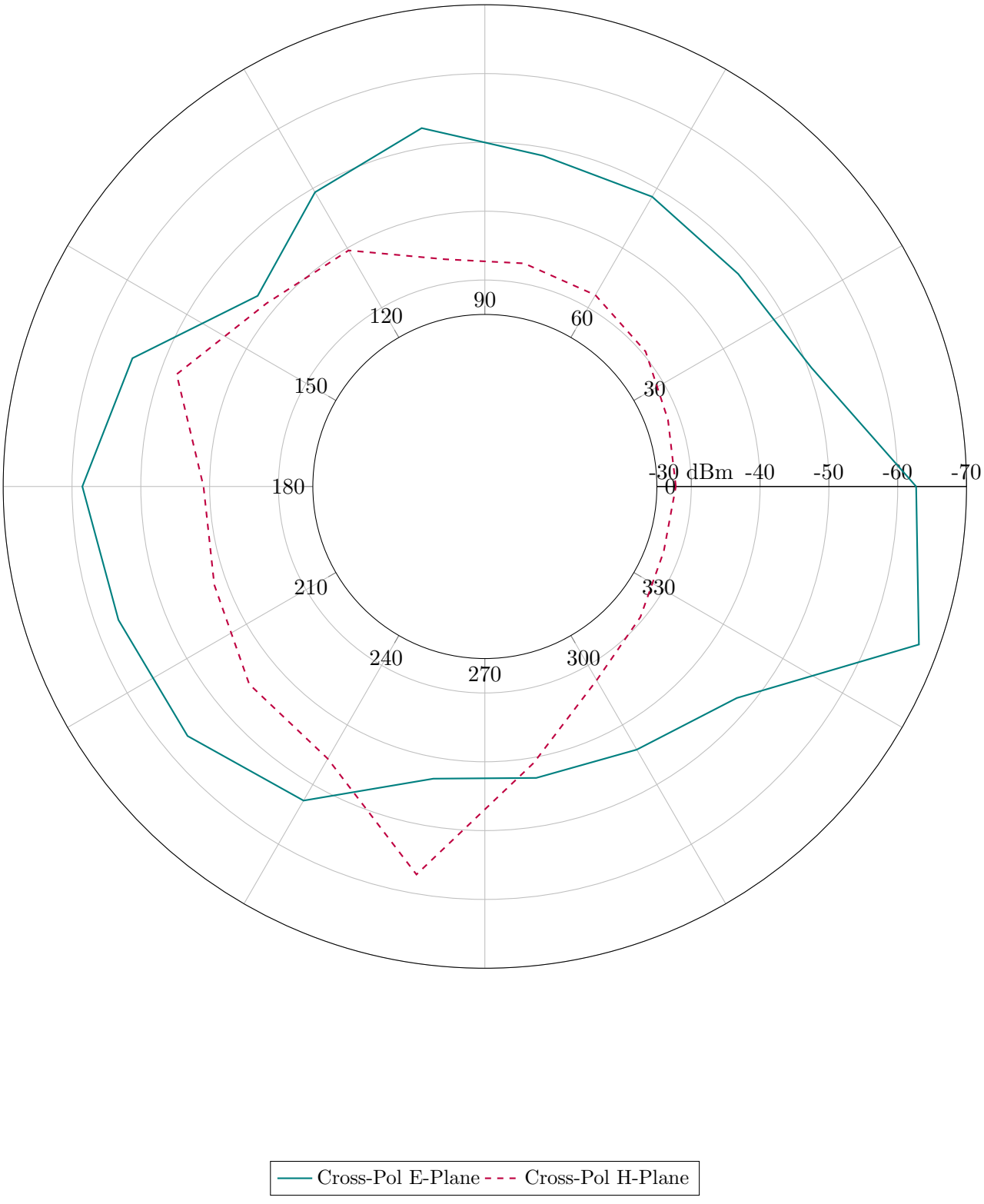


Figure 4: 2D Polar radiation pattern plot for Cross-Polarized E-plane and H-plane configurations.

6 Part E: Half-Power Beamwidth (HPBW) and Directivity Estimation

Half-Power Beamwidth (HPBW) Determination:

1. **E-Plane Co-Polarization:**
   - Peak Received Power ($P_{\text{max, E}}$): -27.34 dBm (at $\theta = 10^\circ$).
   - -3 dB Threshold Level: $-27.34 - 3.0 = \mathbf{-30.34}$ dBm.
   - The main lobe angular span above -30.34 dBm extends from $\theta = 330^\circ$ (-30°) to $\theta = 40^\circ$.
   - $\theta_E = 40^\circ - (-30^\circ) = \mathbf{70.0}^\circ = \mathbf{1.2217}$ rad.
2. **H-Plane Co-Polarization:**
   - Peak Received Power ($P_{\text{max, H}}$): -38.49 dBm (at $\theta = 320^\circ$).
   - -3 dB Threshold Level: $-38.49 - 3.0 = \mathbf{-41.49}$ dBm.
   - Main lobe angular span above -41.49 dBm extends over a broad sector of 80.0°.
   - $\theta_H = \mathbf{80.0}^\circ = \mathbf{1.3963}$ rad.

Directivity (D) Estimation: Using the principal plane beamwidths in radians:

$$D \approx \frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{(1.2217) \times (1.3963)} = \frac{12.5664}{1.7059} = \mathbf{7.37} \text{ (Linear)} \tag{13}$$

Converting to decibels relative to isotropic (dBi):

$$D \text{ (dBi)} = 10 \log_{10}(7.37) = \mathbf{8.67} \text{ dBi} \tag{14}$$

7 Summary of Experimental Results

Table 5: Principal Measured and Derived Antenna Parameters

Parameter / Quantity	**Symbol**	**Measured / Calculated Value**
Monopole Physical Length	L	37.5 cm = 0.375 m
Theoretical Resonant Frequency	f_r	200.0 MHz
Minimum VSWR (at 200 MHz)	VSWR_{min}	1.073
Reflection Coefficient (at 200 MHz)	$\lvert\Gamma\rvert$	0.0352
Return Loss (at 200 MHz)	RL	29.12 dB
E-Plane Half-Power Beamwidth	θ_E	70.0° (1.2217 rad)
H-Plane Half-Power Beamwidth	θ_H	80.0° (1.3963 rad)
Estimated Directivity (Linear)	D	7.37
Estimated Directivity (dBi)	D (dBi)	8.67 dBi
Distance Power Decay Slope	$\text{Slope}_{20 \log R}$	-1.66 (vs. ideal -1.0)

8 Conclusions and Discussion

In this experiment, the performance characteristics of a monopole antenna were experimentally evaluated:

1. **Resonance and Impedance Matching:** The 37.5 cm monopole exhibited near-perfect impedance matching at $f = 200$ MHz (VSWR = 1.073, RL = 29.12 dB), matching the theoretical quarter-wave requirement ($\lambda/4 = 0.375$ m). Off-resonance at 30 MHz, VSWR rose to 1.792.

2. **Distance Dependence:** Received power decreased monotonically with separation distance R. The regression fit against $20 \log_{10} R$ yielded a slope of -1.66, reflecting free-space expansion combined with multipath floor reflections.

3. **Radiation Beam Patterns:**

   - **Co-Polarization:** The E-plane pattern showed a directive main lobe centered near 10°, while the H-plane pattern exhibited a broader, quasi-omnidirectional distribution.
   - **Cross-Polarization:** Rotating the antenna polarization by 90° led to significant signal suppression in the principal lobe direction, proving high polarization selectivity.

4. **Beamwidth and Directivity:** Half-power beamwidths were measured as $\theta_E = 70.0^\circ$ and $\theta_H = 80.0^\circ$, yielding an estimated directivity of $D = 7.37$ (8.67 dBi).